‘Guided Unsupervised Segmentation via Cross-Model Prompting’

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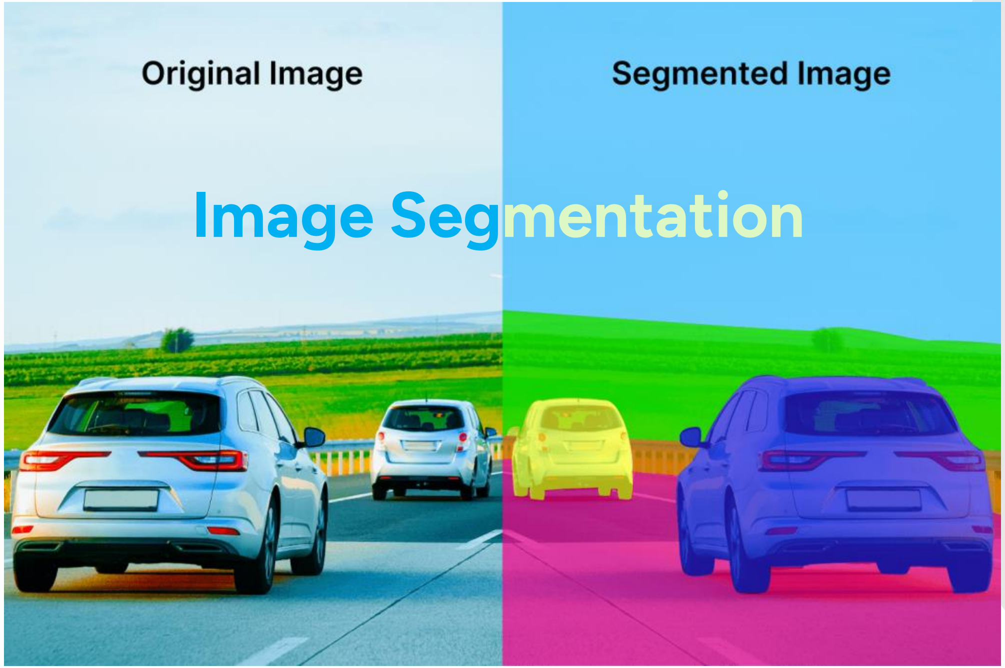
Table of contents

Literature Review

SL No.	Name	Year	Link
1	OneFormer: One Transformer to Rule Universal Image Segmentation	2022	Link
2	SegGPT: Segmenting Everything In Context	2023	Link
3	Attention Is All You Need	2017	Link

01 Introduction



02 Challenges

Real-Time Processing Constraints

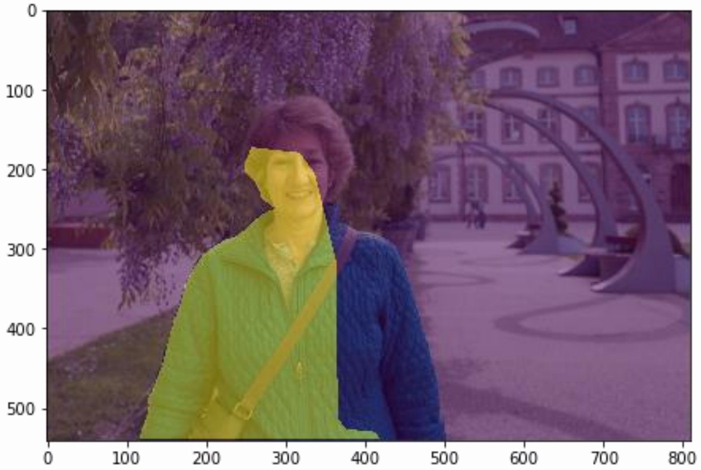
Lack of Contextual Insight

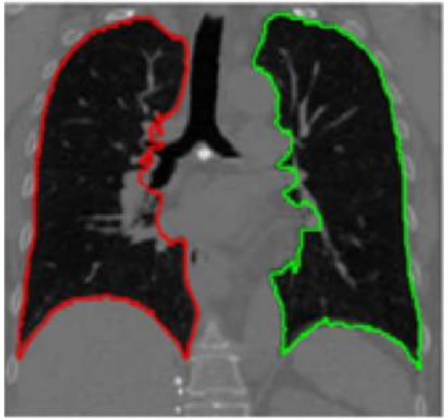
Boundary Detection Challenges

Inconsistent Precision

Limited Data Annotations

Generalization Issues





02 Challenges

Generalization Issues

Limited Data Annotations

Inconsistent Precision

Purpose of the Research:

Unify and Enhance

Enhance **multi-task** segmentation for precision

Unsupervised Segmentation

Precisely Segment with least amount of annotated data and human intervention

Improve Contextual Understanding

Combine multi-task **efficiency** with **deep** contextual understanding.

" Understanding vision and building visual systems is really understanding intelligence"

—Fei-Fei Li

The 100 Most Influential People in AI 2023 | Time

03 Objetives

- Develop Unified Segmentation Framework OneFormer
- Enhance Contextual Understanding with SegGPT
- Optimize Real World Applications

04 Methodology

S1
Input Image Insertion and segment with OneFormer

S2
Object Identification using OneFormer

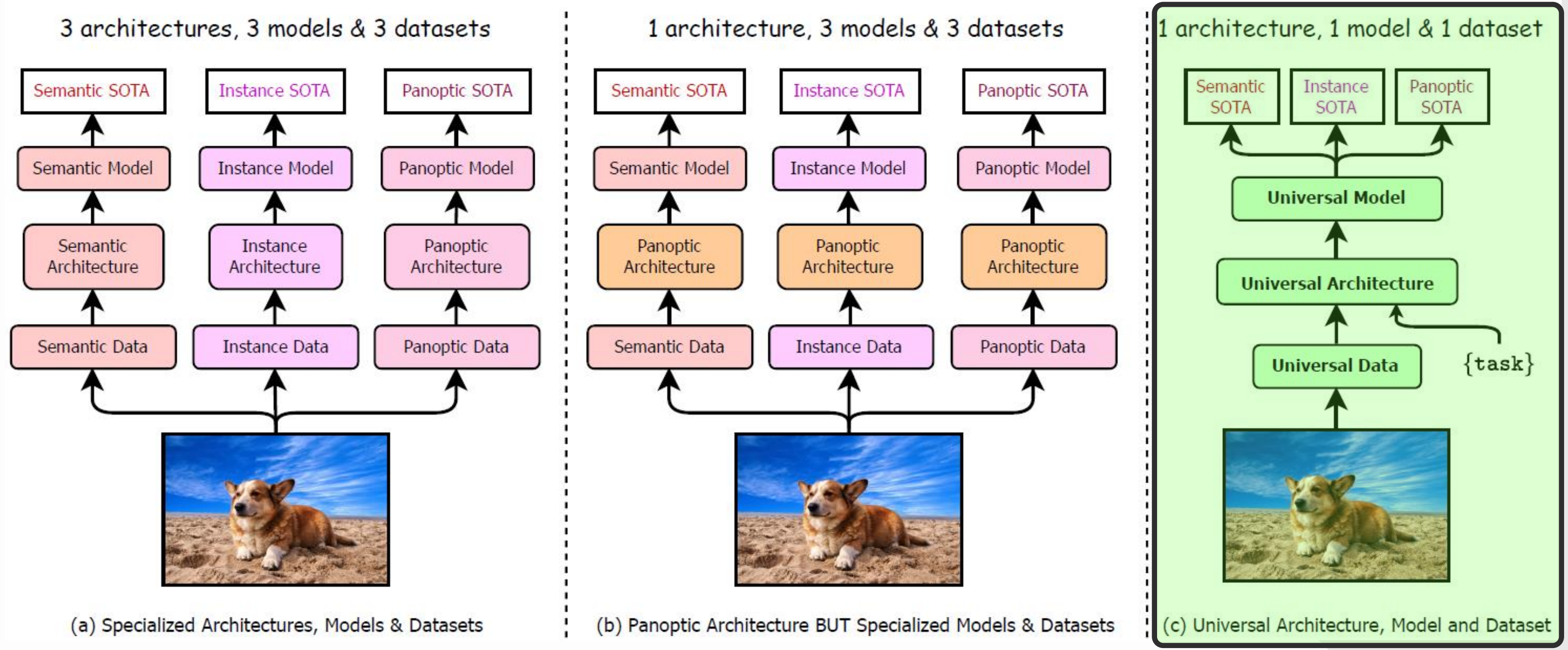
S3
Multi-task Segmentation with Object and Region Labels from OneFormer

S4
Contextual Analysis on Segmented Output with SegGPT

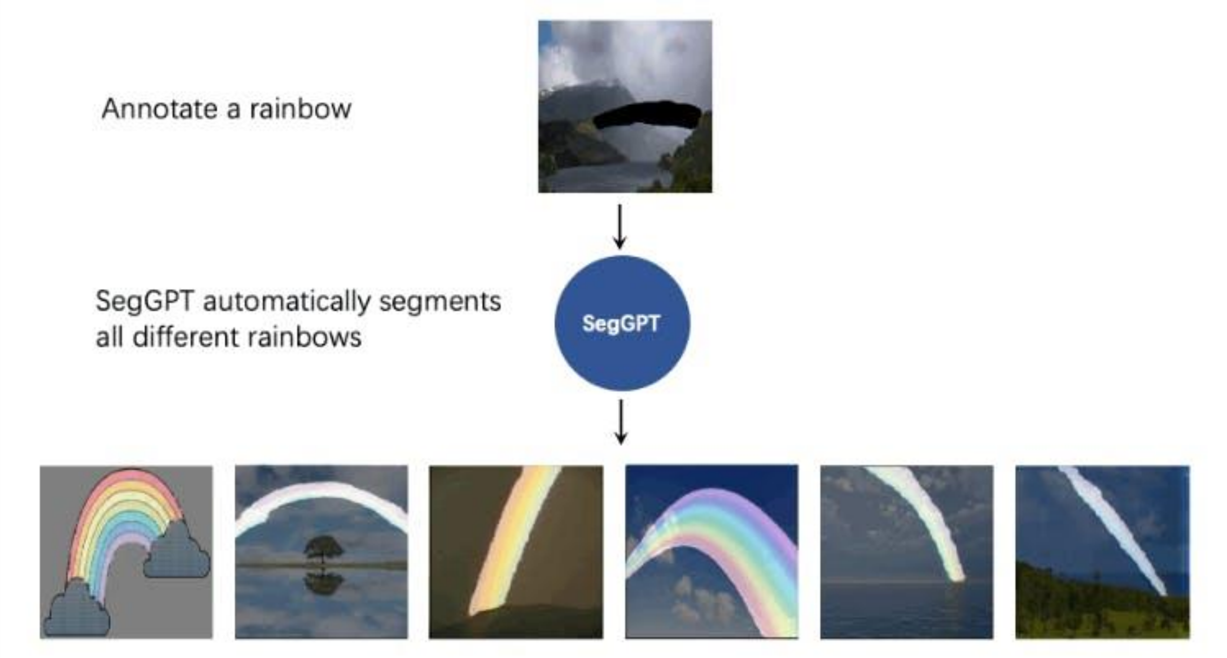
S5
Contextual Refinement by SegGPT for Enhanced Segmentation

S6
Real-Time Optimization for Faster Processing and Efficiency

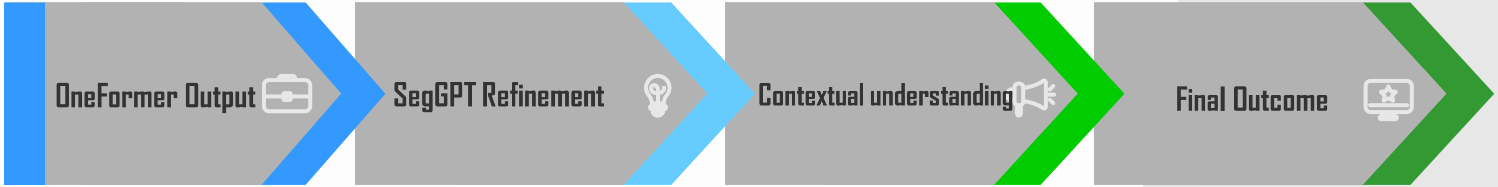
4.1 OneFormer for Initial Segmentation



4.2 SegGPT for Contextual Refinement



How it works



Combine Two SOTA Models

Create a Generalized Model

Leverage Labeled Datasets

Dataset

Brain Tumor Image DataSet : Semantic Segmentation

122 | Code | Download

Data Card

Brain Tumor Image DataSet: Semantic Segmentation

The **Tumor Segmentation Dataset** is designed specifically for the **TumorSeg Computer Vision Project**, which focuses on **Semantic Segmentation**. The project aims to identify tumor regions accurately within **Medical Images** using advanced techniques.

Details of the Dataset:

Project Type: Semantic Segmentation

Subject: Tumor

Classes:

- Tumor (Class 1)
- Non-Tumor (Class 0)

License

Expected update frequency

Never

Tags

Cancer | Computer Vision | Deep Learning | Object Detection | Image Segmentation

Dataset Split

TRAIN SET 70%	VALID SET 20%	TEST SET 10%
1502 Images	429 Images	215 Images

Preprocessing
Auto-Orient: Applied
Resize: Stretch to 640x640

Augmentations No augmentations were applied.

Ref: Brain Tumor Image DataSet: Semantic Segmentation

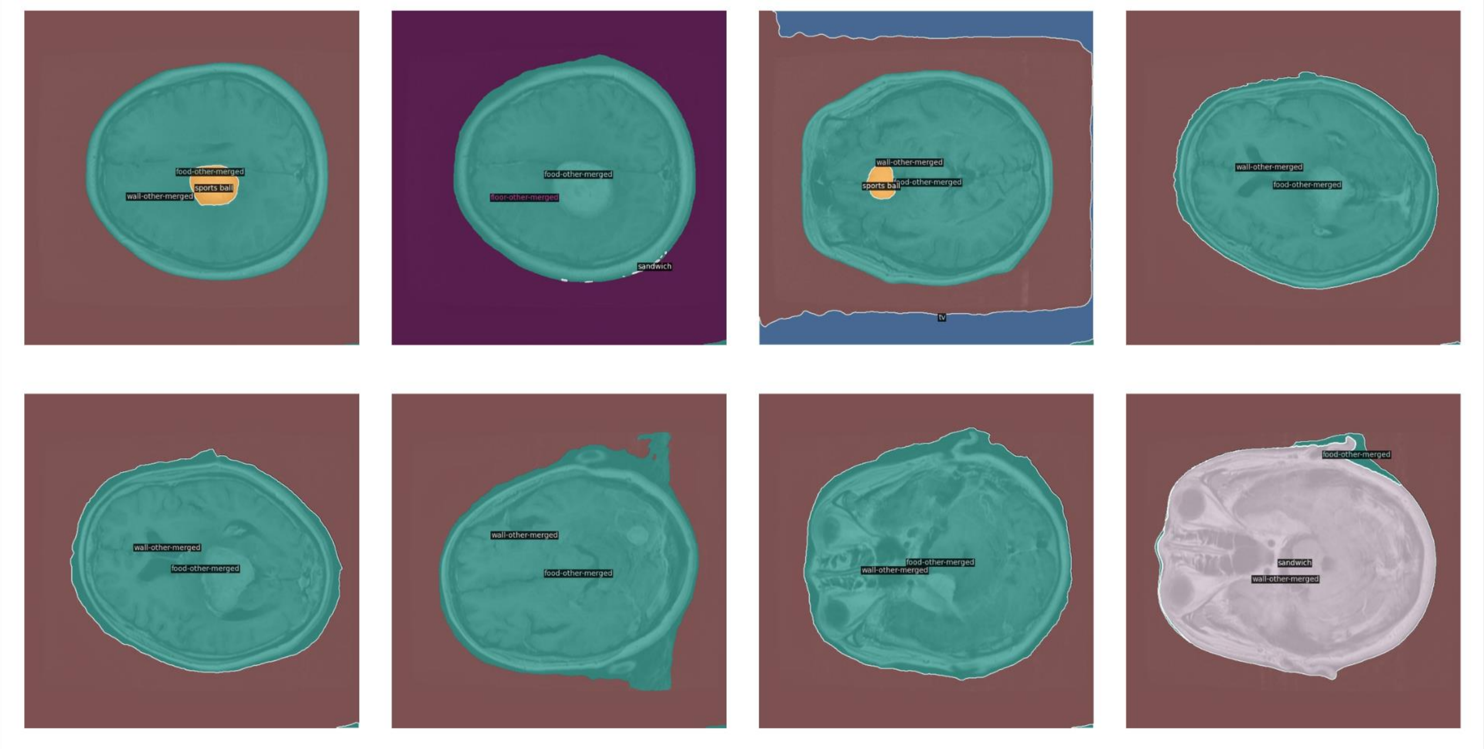


Figure: Segmentation generated by Oneformer

Semantic Segmentation

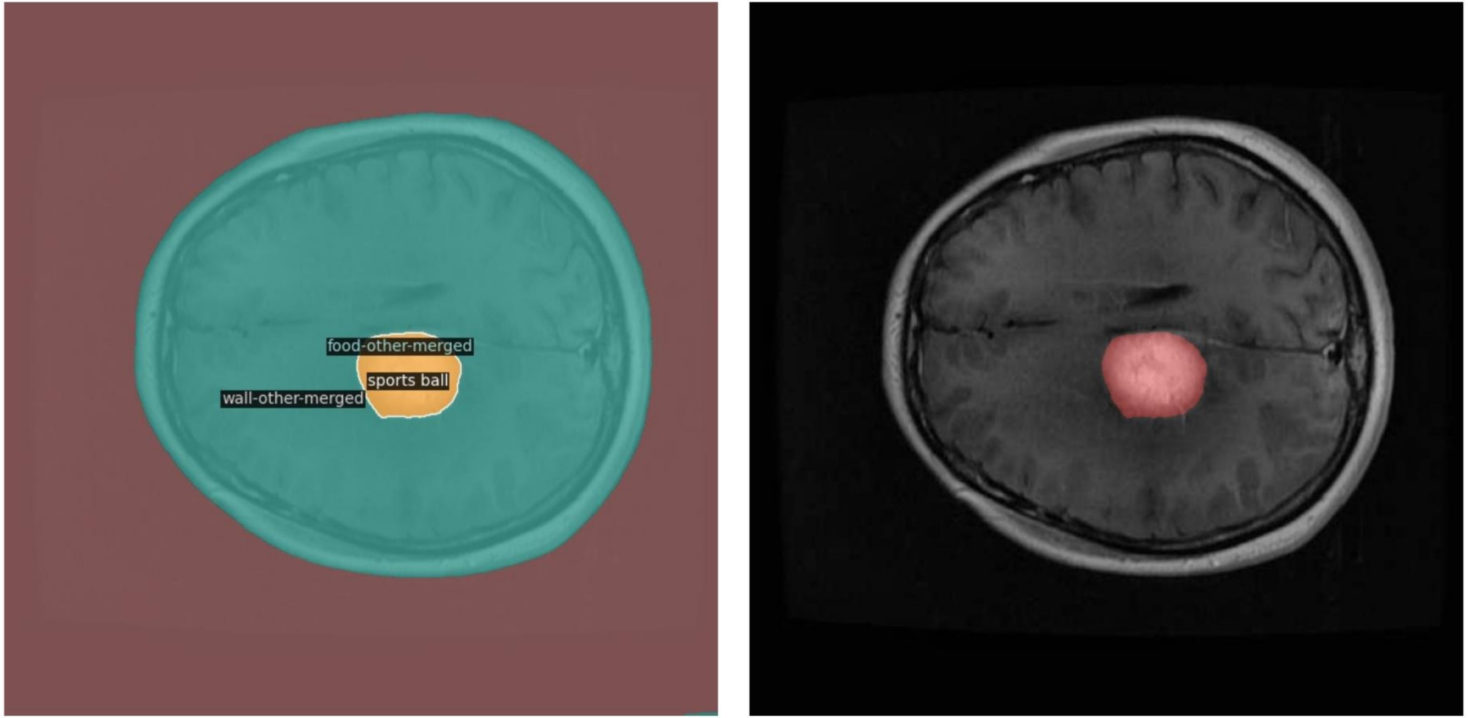


Figure: Predicted Segmentation by OneFormer with Default Label

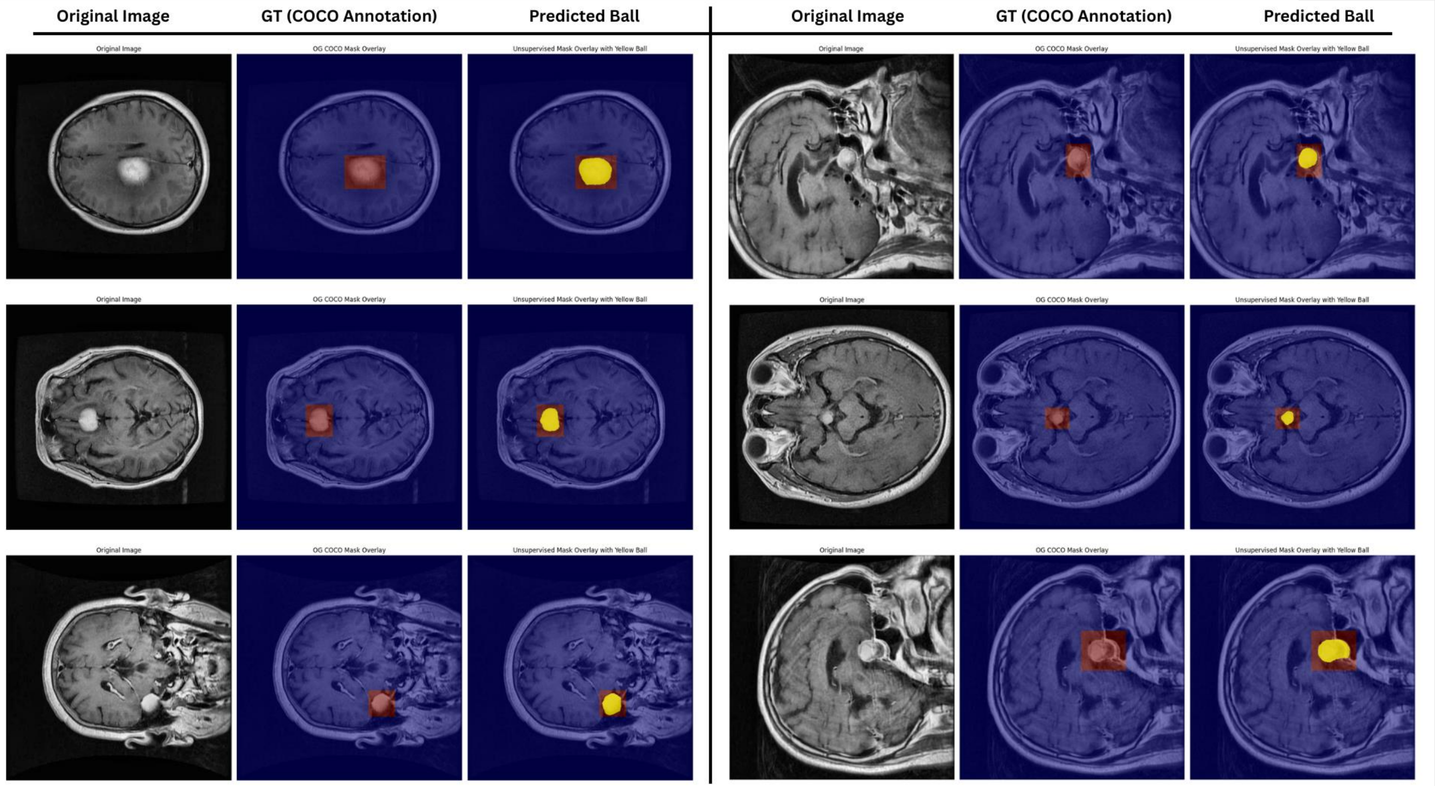


Figure: Ground Truth and Predicted Mask overlayed on top or Original Image for Reference

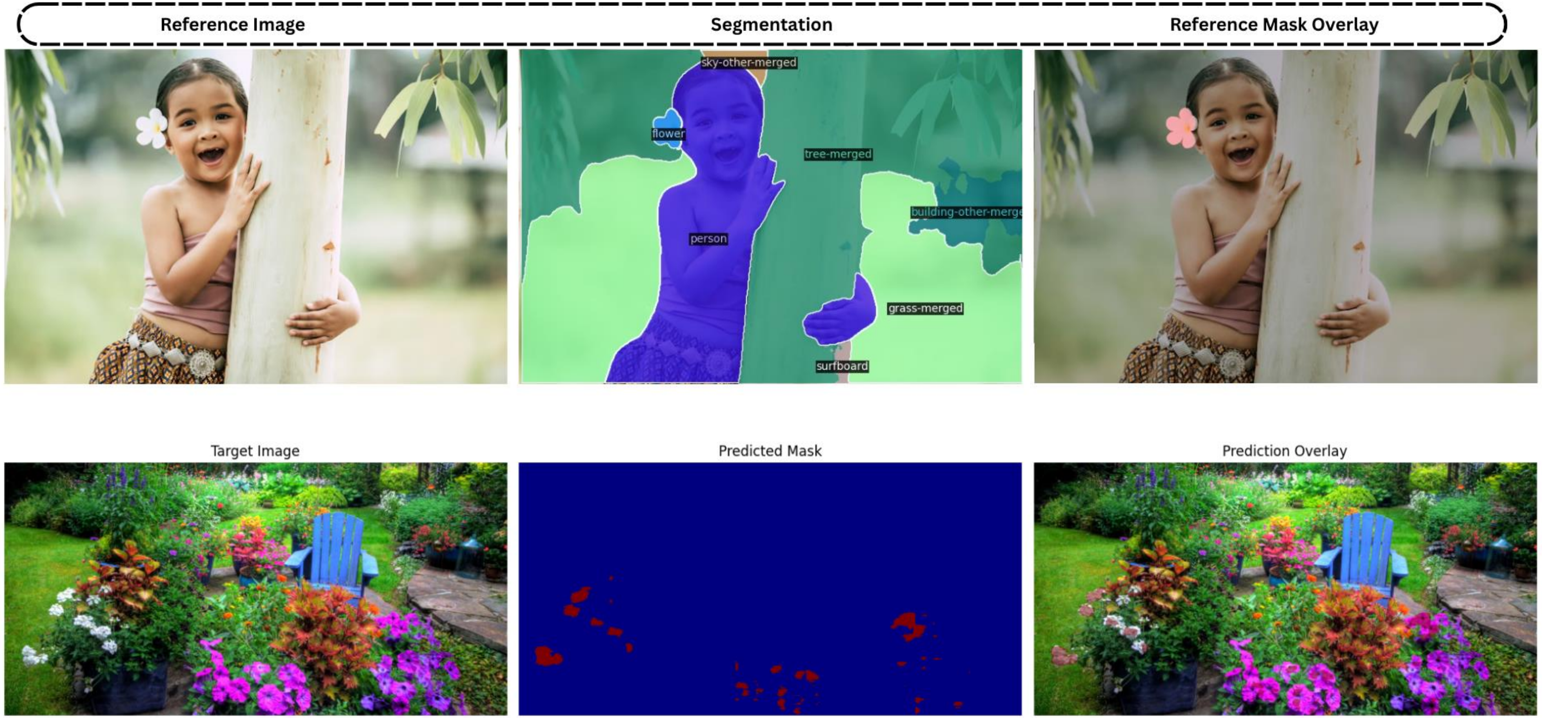
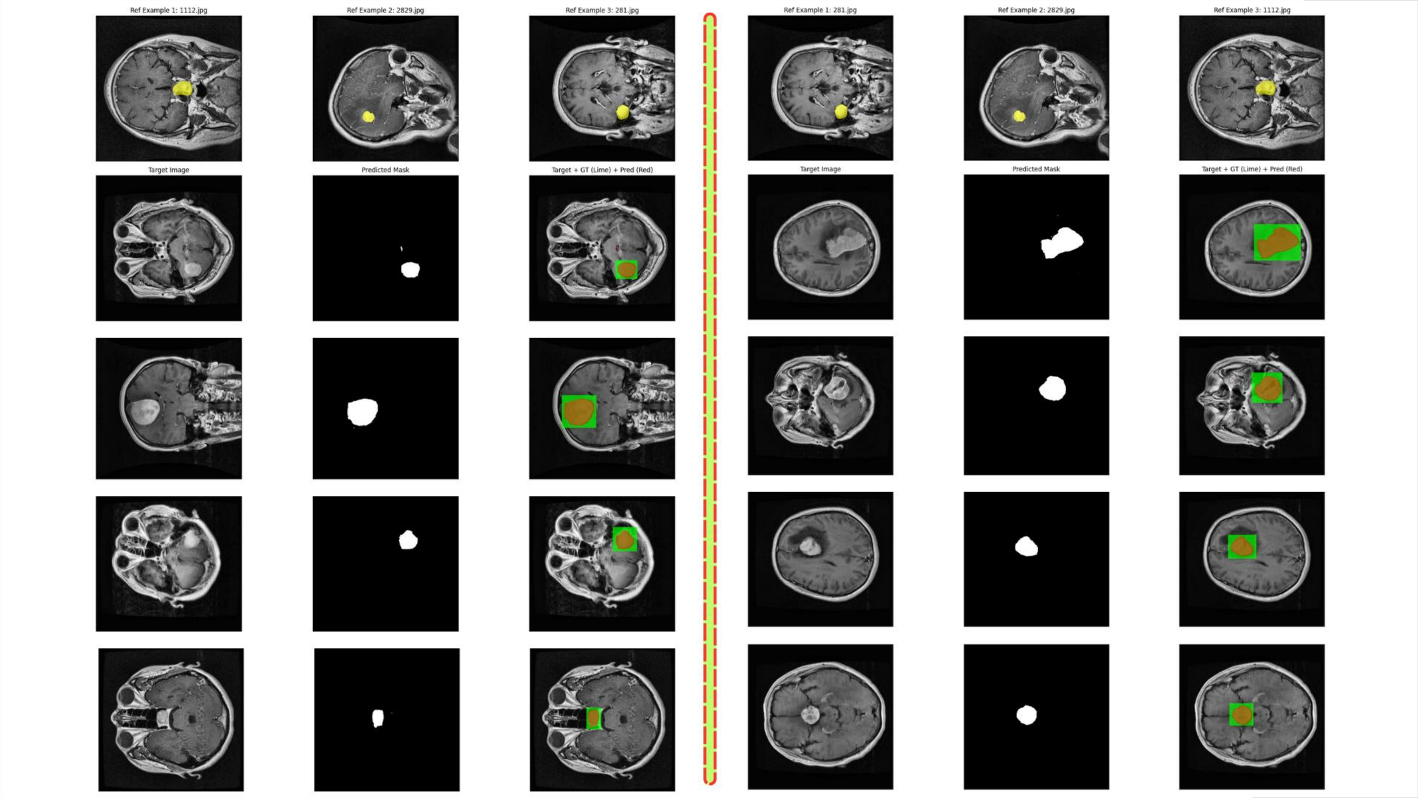


Figure: Reference image and reference mask passed into SegGPT for Contextual understanding


Ref Example 1: 1112.jpg
Ref Example 2: 2829.jpg
Ref Example 3: 281.jpg
Target Image
Predicted Mask
Target + GT (Lime) + Pred (Red)
Ref Example 1: 281.jpg
Ref Example 2: 2829.jpg
Ref Example 3: 1112.jpg
Target Image
Predicted Mask
Target + GT (Lime) + Pred (Red)

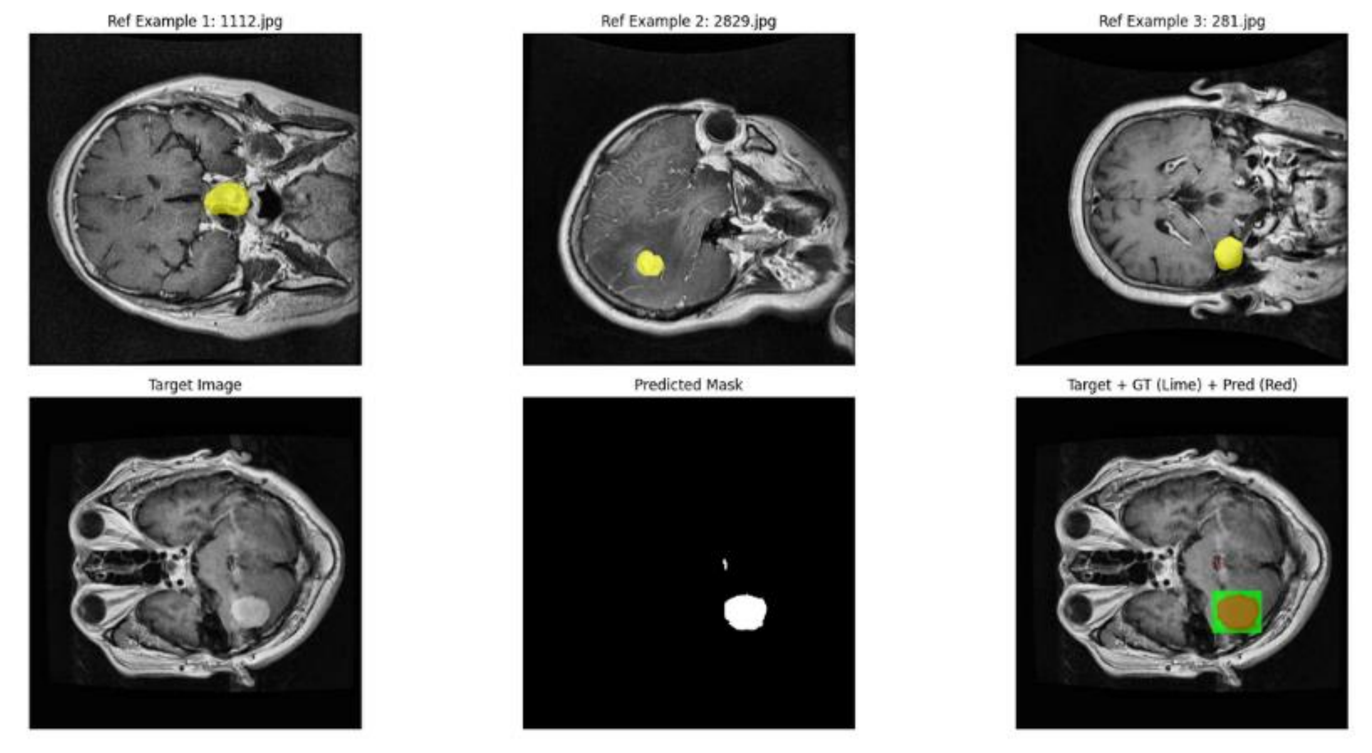
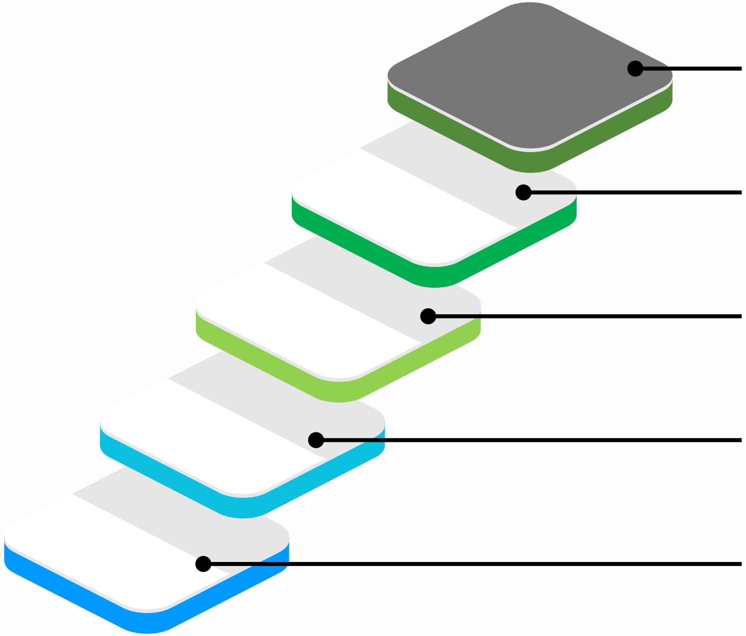


Figure: Predicted Segmentation (using SegGPT) Compared to Ground Truth

Why It Should Work



- Powerful Base Models
- Complementary Strengths
- Labeled Context
- Distillation Potential
- Domain Versatility

Challenges

High Computational Cost

Demands Significant Resources

Context Confusion

SegGPT may struggle with OneFormer's outputs

Limited Generalization

Too much dependency on specific example

VRAM Constraints

Hard to handle large images

05 Results

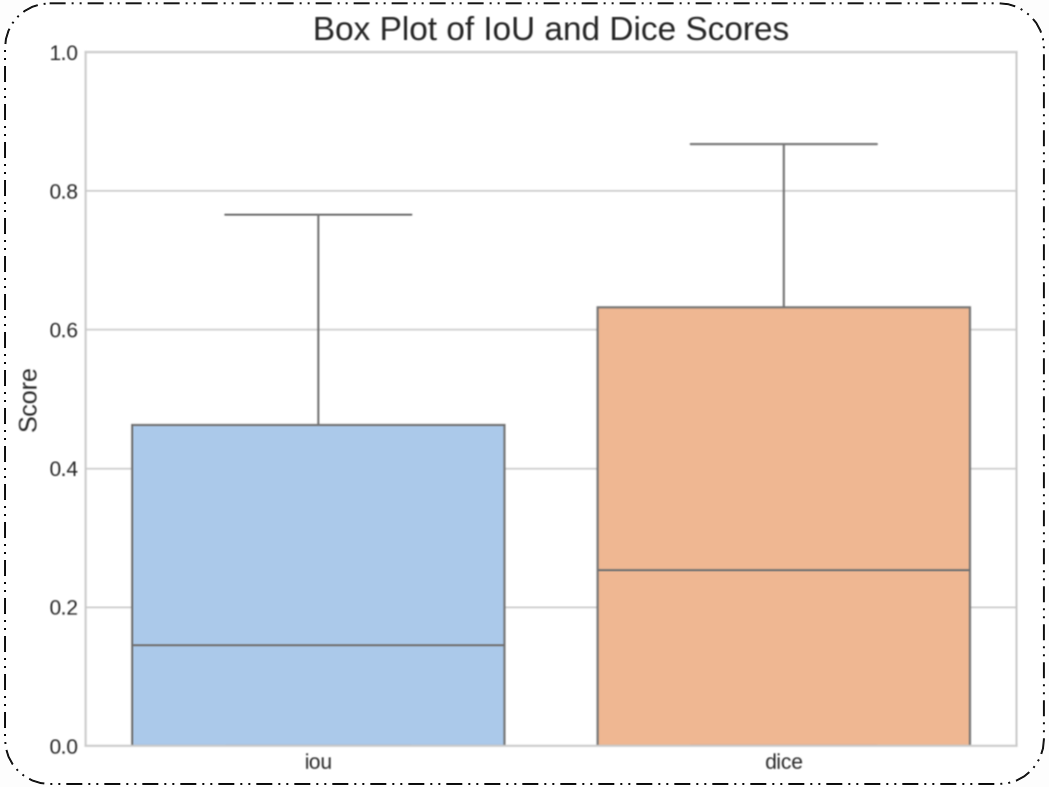


Figure: Comparison of IoU and Dice Score

05 Results

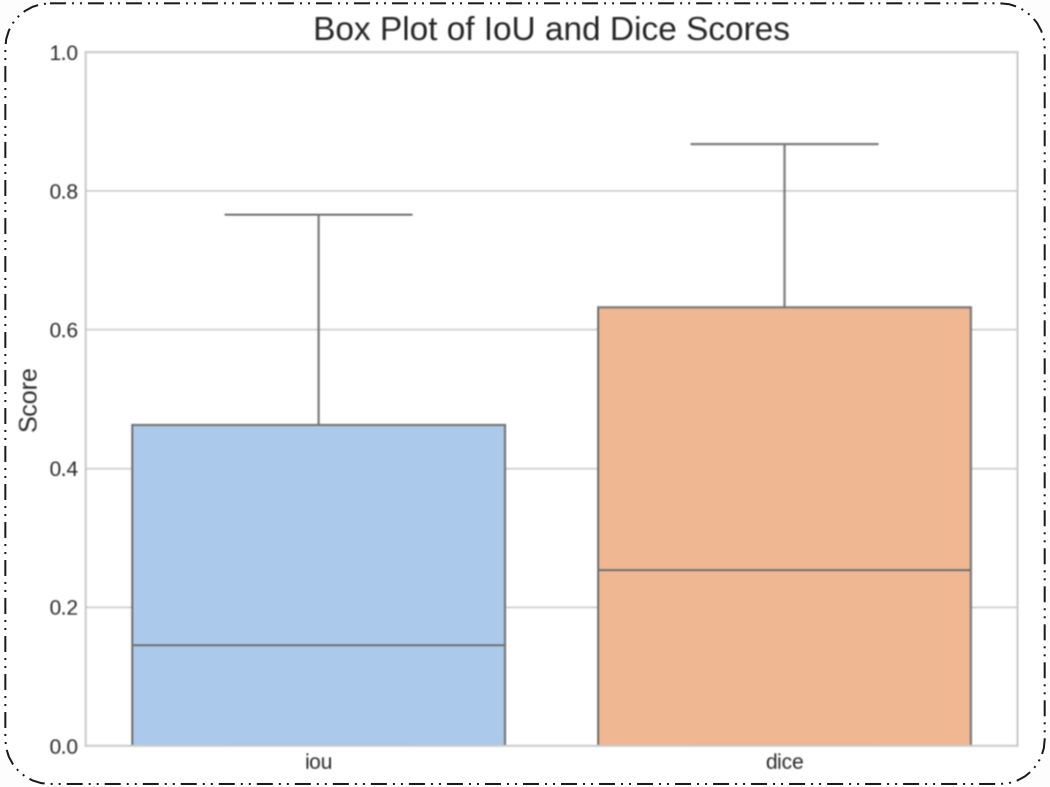


Figure: Comparison of IoU and Dice Score

Best Dice Score: 87%
IoU Score: 77%
*Overlap of predicted mask(GT)

06 Applications of our approach

Autonomous Driving

Defense Systems

Medical Imaging

Augmented Reality

Agricultural Monitoring

07 Conclusions

Enhanced Segmentation Accuracy

Context Aware Refinement

Real-Time Application Readiness

Foundation for Future Work

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Thanks!

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